

YUZHNO-KURILSK, Russian Federation

WMO: 321650

Lat: 44.0229N

Lon: 145.8630E

Elev: 49

StdP: 100.74

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-11.2	-9.7	-16.8	0.9	-9.8	-15.2	1.0	-8.7	17.1	-4.4	15.6	-4.2	4.5	320	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.1	21.3	18.8	20.0	18.1	19.0	17.5	19.6	20.5	18.7	19.5	17.9	18.7	3.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.1	20.1	18.4	13.3	19.2	17.5	12.6	18.4	56.5	20.7	53.4	19.5	50.8	18.8	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	12.2	10.4	-12.4	24.8	2.1	2.4	-13.8	26.5	-15.0	27.9	-16.2	29.2	-17.7	31.0
			-13.2	20.9	2.0	1.6	-14.7	22.1	-15.9	23.0	-17.0	23.9	-18.4	25.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-4.1	-5.1	-2.0	2.0	5.7	9.0	13.1	16.1	15.5	11.0	4.8	-1.1
	DBStd	7.69	2.57	2.92	2.65	2.44	2.68	2.39	2.49	2.43	2.10	2.63	3.38	2.74
	HDD10.0	2177	437	422	373	240	137	46	3	0	0	20	157	343
	HDD18.3	4705	695	655	631	490	392	280	162	77	88	228	405	601
	CDD10.0	525	0	0	0	0	3	16	100	189	165	50	2	0
	CDD18.3	12	0	0	0	0	0	0	1	8	3	0	0	0
	CDH23.3	12	0	0	0	0	0	0	1	9	2	0	0	0
	CDH26.7	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	4.4	5.5	5.2	4.9	4.3	3.8	3.3	3.0	3.3	4.1	5.0	5.4	5.5
Precipitation	PrecAvg	1238	58	42	68	89	129	128	125	122	170	133	102	71
	PrecMax	1623	133	94	130	149	330	251	241	205	406	277	196	125
	PrecMin	921	26	10	18	30	46	29	39	43	78	12	25	24
	PrecStd	180	28	22	30	31	67	59	52	55	68	67	43	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	2.4	6.1	11.9	16.1	18.2	21.9	24.6	22.5	17.9	13.2	7.9
		MCWB	2.2	0.9	2.6	7.1	10.6	14.7	18.5	20.2	19.1	14.0	11.8	7.0
	2%	DB	1.6	1.0	3.6	8.9	13.0	15.2	19.7	21.8	20.6	16.5	12.1	6.1
		MCWB	0.5	-0.2	1.6	5.3	9.3	12.8	17.8	19.4	18.4	14.2	10.7	4.9
	5%	DB	0.7	0.0	2.2	6.8	11.0	13.6	17.9	20.4	19.5	15.6	11.1	4.7
		MCWB	-0.5	-1.0	0.6	4.1	8.2	12.0	16.7	18.6	17.5	13.5	9.5	3.3
	10%	DB	-0.2	-0.7	1.3	5.3	9.4	12.2	16.6	19.3	18.6	14.7	10.0	3.1
		MCWB	-1.4	-1.7	-0.1	3.4	7.5	11.1	15.7	18.1	16.8	12.8	8.4	1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	1.4	3.5	7.5	11.6	15.3	19.4	21.3	20.7	16.2	12.6	7.2
		MCDB	2.8	1.9	4.7	10.9	14.4	17.8	20.8	23.1	21.4	16.8	12.9	7.8
	2%	WB	0.8	0.1	2.0	5.8	9.9	13.3	18.0	20.1	19.3	15.0	11.2	5.2
		MCDB	1.3	0.6	3.1	7.9	11.9	14.5	19.4	21.2	19.8	15.8	11.8	5.9
	5%	WB	-0.3	-0.8	1.0	4.6	8.8	12.3	16.8	19.1	18.3	14.1	9.8	3.6
		MCDB	0.4	-0.2	1.9	6.2	10.3	13.3	17.6	19.9	19.0	15.1	10.8	4.6
	10%	WB	-1.4	-1.6	0.2	3.8	7.9	11.3	15.8	18.3	17.3	13.1	8.6	1.7
		MCDB	-0.3	-0.8	1.0	5.0	9.2	12.0	16.5	19.1	18.2	14.4	9.9	2.9
Mean Daily Temperature Range	5% DB	MDBR	3.6	4.3	4.1	4.4	4.4	3.5	3.3	3.1	3.5	4.1	4.3	3.6
		MCDBR	4.6	5.0	5.1	7.4	7.9	6.3	6.2	5.2	4.6	4.6	5.5	5.5
		MCWBR	4.7	4.7	4.2	4.8	4.9	4.3	4.4	3.5	3.6	4.3	5.8	6.2
	5% WB	MCDBR	4.6	4.9	4.8	6.5	6.8	5.5	5.7	4.5	3.7	4.0	5.5	5.6
		MCWBR	5.0	4.9	4.3	4.8	4.7	4.1	4.4	3.4	3.5	4.4	6.3	6.6
Clear-Sky Solar Irradiance	taub	0.338	0.371	0.440	0.501	0.512	0.486	0.467	0.452	0.407	0.399	0.382	0.354	
	taud	2.186	2.043	1.906	1.866	1.936	2.114	2.240	2.301	2.371	2.238	2.179	2.148	
	Ebn at Noon	753	789	777	765	772	796	807	806	815	757	692	684	
	Edh at Noon	94	129	169	191	184	154	135	123	106	106	95	89	
All-Sky Solar Radiation	RadAvg	1.47	2.42	3.59	4.46	4.65	4.75	4.43	3.99	3.54	2.64	1.61	1.23	
	RadStd	0.12	0.19	0.27	0.33	0.40	0.40	0.45	0.41	0.28	0.13	0.14	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.29	N/A	N/A	N/A	N/A	N/A	N/A	-79	N/A	N/A

Nomenclature: See separate page